

Multiple Choice (10 marks)

1. What is the typical body composition of a normal weight young adult man?
 - (a) Fat = 17%, Fat Free Mass=83%, Intracellular Fluid= 40%, Extracellular Fluid= 20%
 - (b) Fat = 83%, Fat Free Mass=40%, Intracellular Fluid= 17%, Extracellular Fluid= 20%
 - (c) Fat = 20%, Fat Free Mass=17%, Intracellular Fluid= 40%, Extracellular Fluid= 83%
 - (d) Fat = 40%, Fat Free Mass=20%, Intracellular Fluid= 17%, Extracellular Fluid= 83%
2. The vitamins which are lost most readily during cooking are:
 - (a) vitamin E and niacin
 - (b) vitamin D and vitamin E
 - (c) riboflavin and vitamin C
 - (d) vitamin C and thiamin
3. Which foods tend to be consumed in lower quantities in Wales and Scotland (as of 2020)?
 - (a) Meat
 - (b) Confectionary
 - (c) Fruits and vegetables
 - (d) Potatoes
4. Which of the following vitamins provides the coenzyme for carboxylation reactions?
 - (a) Biotin
 - (b) Niacin
 - (c) Riboflavin
 - (d) Thiamin
5. What is the estimated global prevalence of untreated dental caries?
 - (a) 20%
 - (b) 40%
 - (c) 60%
 - (d) 80%

True / False (10 marks)

Answer True or False

6. True or False: High intakes of foods containing phytic acid can reduce the absorption of potassium in the body.
7. True or False: Recumbent length is a common anthropometric measure used to assess the length of infants.
8. True or False: HIV infection rates in Europe have been significantly decreasing in the past few years.
9. True or False: Wilson's disease is an inborn error of metabolism that causes zinc deficiency in the body.
10. True or False: De novo fatty acid synthesis typically occurs when there is an excess of energy yielding substrates in the body.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of intrauterine growth retardation (IUGR) and explain how it relates to birth weight thresholds for gestational age, including its implications for fetal health.
12. Discuss the role of fat-soluble vitamins in diet and analyze their impact on cardiovascular disease, considering current research and recommendations.

End of Paper